Agenda

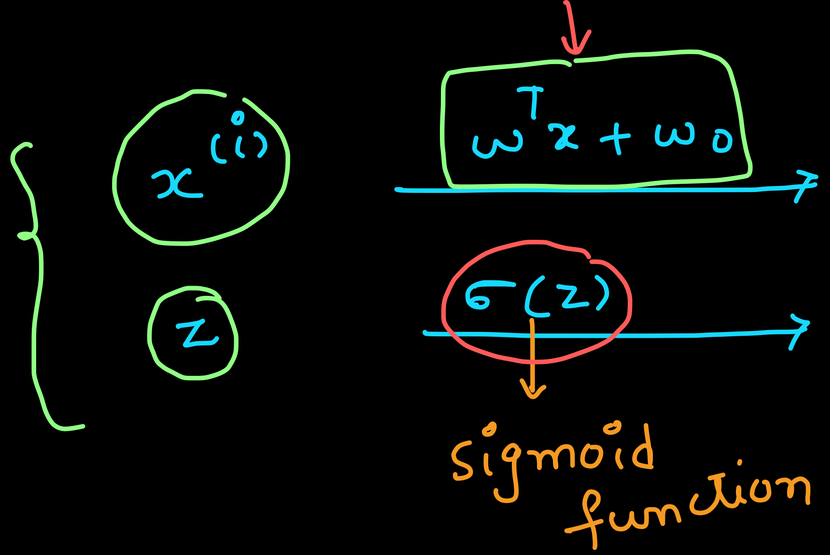
* Derivation $\Rightarrow$ $\frac{\partial L}{\partial w_j}$
* Hyper parameter Tuning
* Logodds
* Impact of outliers
* Multiclass classification

✓ Assignments

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Recap

Process :



(distance from line)

$(0,1) \Rightarrow \{0,1\}$ $\geq 0.5 \Rightarrow 1$, 0

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$\Rightarrow$

1. $z \to \infty$ v. large +ve
   $\sigma(z) \to 1$
2. $z \to -\infty$ v. large -ve
   $\sigma(z) \simeq 0$
3. $z \to 0$
   $\sigma(z) = 0.5$

Property: *** $\sigma(z^i) = p(y^i = 1 \mid x^i)$

$\Rightarrow$ It will give you the prob of the data point belongs to class 1

Loss Function

loglos

$$L = \frac{-1}{n} \sum_{i=1}^{n} \left[y^i \log \hat{y}^i + (1 - y^i) \log (1 - \hat{y}^i) \right]$$

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$$L = \frac{-1}{n} \sum_{i=1}^{n} \left[y^i \log \hat{y}^i + (1 - y^i) \log (1 - \hat{y}^i) \right] + \lambda \sum_{j=1}^{d} w_j^2$$

log loss

$$2\lambda \sum_{j=1}^{d} w_j$$

Take one data point

$$L = -\left[y \log \hat{y} + (1 - y) \log (1 - \hat{y}) \right]$$

$$\hat{y} = \sigma(z) = \sigma(w^T x + w_0)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{d(\sigma(z))}{dz} = \sigma(z) * (1 - \sigma(z))$$

$$= p$$

$$= \sigma(z) * (1 - \sigma(z))$$

$$\boxed{\frac{d\hat{y}}{dz} = \hat{y} * (1 - \hat{y})}$$

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$$f(x) = \frac{1}{1+e^{-x}} \rightarrow \text{Sigmoid}$$

($1 \rightarrow f(x)$, $1+e^{-x} \rightarrow g(x)$)

$$f'(x) = \frac{(1+e^{-x})\cdot\frac{d}{dx}(1) - 1\cdot\frac{d}{dx}(1+e^{-x})}{(1+e^{-x})^2}$$

$$= \frac{0 - \left[1\cdot 0 + \frac{d}{dx}e^{-x}\right]}{(1+e^{-x})^2}$$

$$= \frac{-\left[0 \quad e^{-x}(-1)\right]}{(1+e^{-x})^2} = \frac{e^{-x}}{(1+e^{-x})(1+e^{-x})}$$

Snapshot saved.

$$\Rightarrow \frac{1}{(1+e^{-x})}\left[\frac{e^{-x}}{1+e^{-x}}\right]$$

$$\Rightarrow \frac{1}{1+e^{-x}}\left[1 - \frac{1}{1+e^{-x}}\right]$$

$$\Rightarrow f(x)$$

$$\Rightarrow \boxed{f(x)\ (1-f(x))} \Rightarrow \text{Deep learning}$$

where $f(x)$ is a sigmoid function

$$L = -\left[y\cdot\log\hat{y} + (1-y)\log(1-\hat{y})\right]$$

part A | part B

$$\frac{\partial L}{\partial w_j} = \frac{\partial L_A}{\partial w_j} + \frac{\partial L_B}{\partial w_j}$$

$$\hat{y} = \sigma(w^T x + w_0)$$

$$= \sigma(w_1 x_1 + w_2 x_2 + \cdots + w_j x_j + \cdots + w_d x_d + w_0)$$

chain rule

$$\Rightarrow \frac{\partial L_A}{\partial w_j} = \frac{\partial L_A}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

$$L_A = y \cdot \log \hat{y}$$

$$= y \cdot \frac{1}{\hat{y}} \cdot \hat{y}(1-\hat{y}) \cdot x_j$$

$$\frac{d}{w_j} w_j x_j$$

$$\boxed{\frac{\partial L_A}{\partial w_j} = y(1-\hat{y})\, x_j}$$

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$$\frac{\partial L_B}{\partial w_j} = \boxed{\frac{\partial L_B}{\partial \hat{y}}} \frac{\partial \hat{y}}{\partial z} \cdot \boxed{\frac{\partial z}{\partial w_j}}$$

$$L_B = (1-y)\log(1-\hat{y})$$

$$= (1-y) * \frac{1}{(1-\hat{y})}(-1) \cdot \hat{y}\cdot(1-\hat{y}) \cdot x_j$$

$$\boxed{\frac{\partial L_B}{\partial w_j} = -\hat{y}\,(1-y)\cdot x_j}$$

$$\frac{\partial L}{\partial w_j} = \frac{\partial L_A}{\partial w_j} + \frac{\partial L_B}{\partial w_j}$$

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$$= \underline{\underline{y(1-\hat{y})x_j}} - (1-y)\hat{y}x_j$$

$$= x_j \left[y - \cancel{y\hat{y}} - \hat{y} + \cancel{y\hat{y}} \right]$$

$$= x_j (y - \hat{y})$$

$$\boxed{\frac{\partial L}{\partial w_j} = (\hat{y} - y) \cdot x_j} \quad \checkmark$$

$$\boxed{\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}^i - y^i) x_j^i}$$

Linear reg ⇒ $\frac{\partial L}{\partial w_j} = \frac{2}{n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) x_j^i$

Difference

lin. reg $\hat{y}$ ⇒ $w^T x + w_0 \in \mathbb{R}$

log reg $\hat{y}$ ⇒ $\sigma(w^T x + w_0) \in (0,1)$

GD ↓

$$w_j = w_j - \eta * \left[\frac{1}{n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) x_j^i + 2\lambda \sum_{j=1}^{d} w_j \right]$$

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$$C = \frac{1}{\lambda}$$

$\lambda = 10000 \Rightarrow C = 0.0001$

underfit

$\lambda = 0.0001 \Rightarrow C = 10000$

Hyper parameter tuning

What is the effect of increasing the regularization rate (C) in logistic regression?

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A	The model becomes less prone to overfitting	0%
B	The model's training accuracy increases	0%
C	The model becomes more prone to overfitting	0%
D	The model's test accuracy increases	0%

End Quiz Now

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odds

- odds of Ind = 4 : 1 → 5

odds ratio = prob of winning / prob of lossing = $\frac{4/5}{1/5}$

odds = $\frac{P}{1-P}$

odds of pak = 1 : 4

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logistic reg model predicts =>

Log odds

$$p(y=1 \mid x_i) = p = \sigma(z)$$

$$p = \sigma(w^T x + w_0)$$

$$p = \frac{1}{1+e^{-z}}$$

$$p = \frac{1}{1+\left(\frac{1}{e^z}\right)}$$

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$$p = \frac{1}{\left(1 + \frac{1}{e^z}\right)}$$

$$= \frac{1}{\frac{e^z + 1}{e^z}}$$

$$\boxed{p = \frac{e^z}{1 + e^z}}$$

$$1 - p = 1 - \frac{e^z}{1 + e^z}$$

$$= \frac{1 + e^z - e^z}{1 + e^z}$$

$$\boxed{1 - p = \frac{1}{1 + e^z}}$$

$$\text{odds} = \frac{p}{1-p} = \frac{\frac{e^z}{1+e^z}}{\frac{1}{1+e^z}}$$

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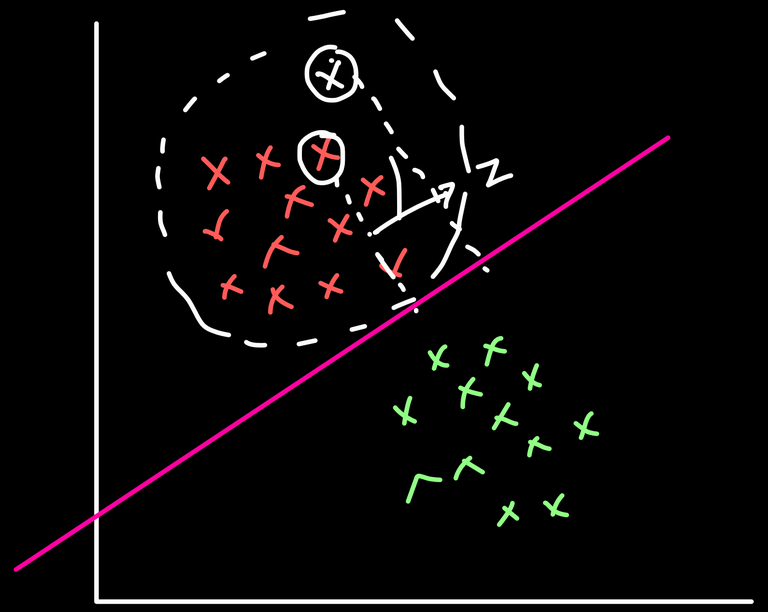
$$\frac{p}{1-p} = e^{z}$$

$$\log_e\left(\frac{p}{1-p}\right) = \log_e e^{z}$$

$$\log_e\left(\frac{p}{1-p}\right) = z$$

$\log_e\left(\frac{p}{1-p}\right)$: odds

logodds



** Distance of line & points gives by logodds

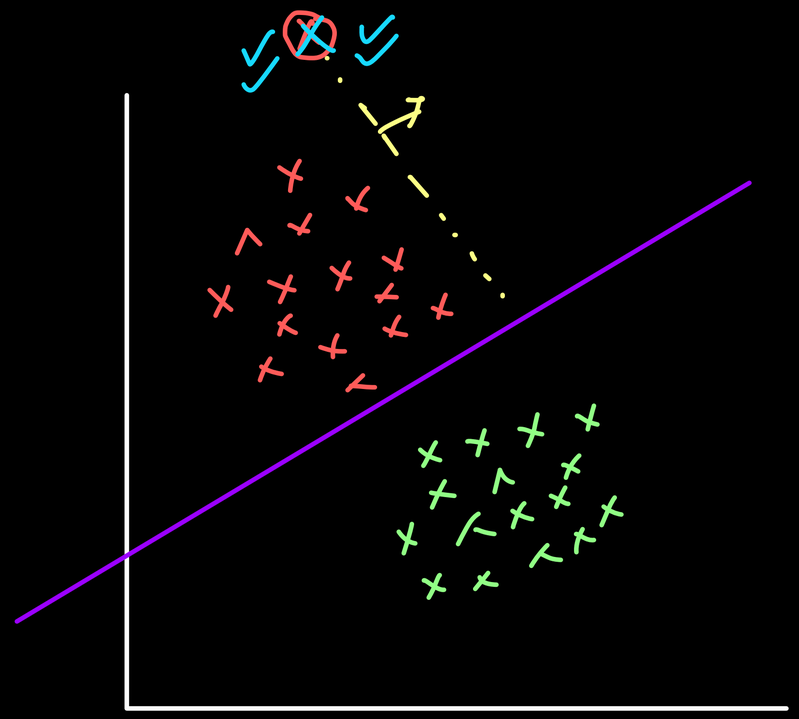
Impact of outliers

case 1

$y = 1$ ✓✓

$$L = -[(y)\log\hat{y} + (1-y)\cdot\log(1-\hat{y})]$$

A B

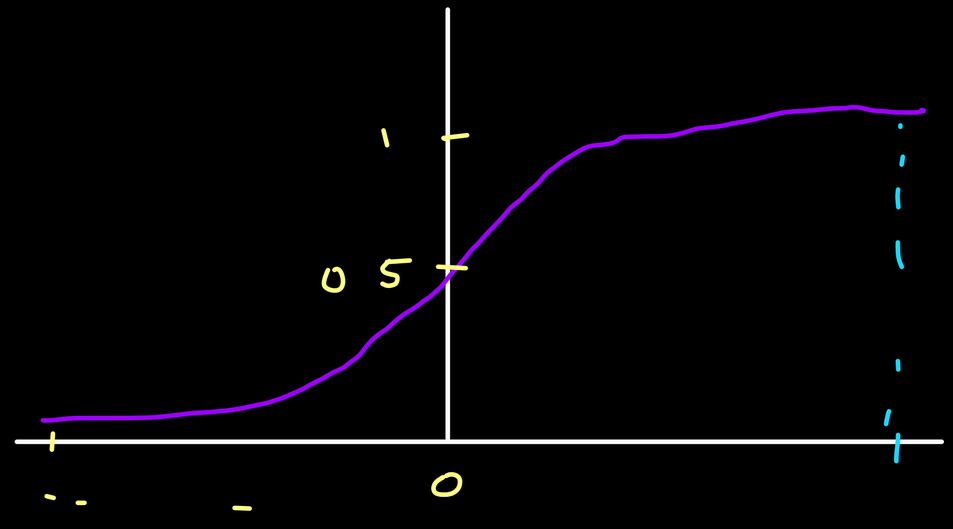


$$L = -\log\hat{y}$$

$\hat{y} = \sigma(z)$

$z \uparrow$

$L = 0$



Case 2

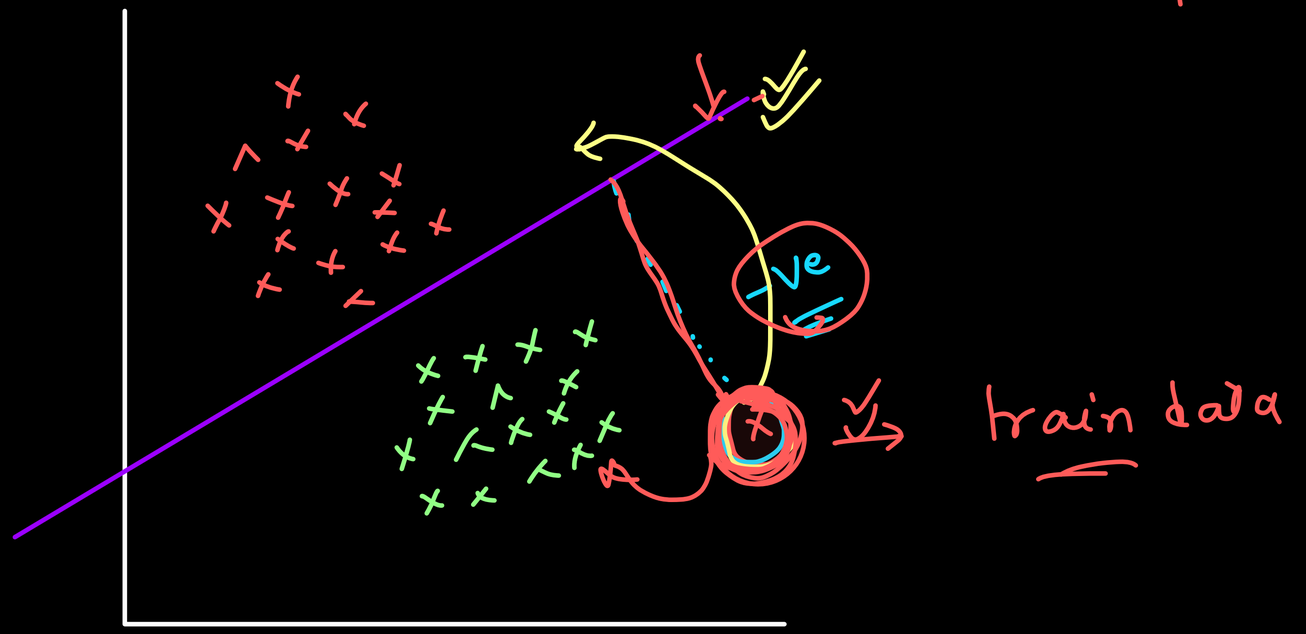
$y = 1$

$L = -\log(\hat{y})$

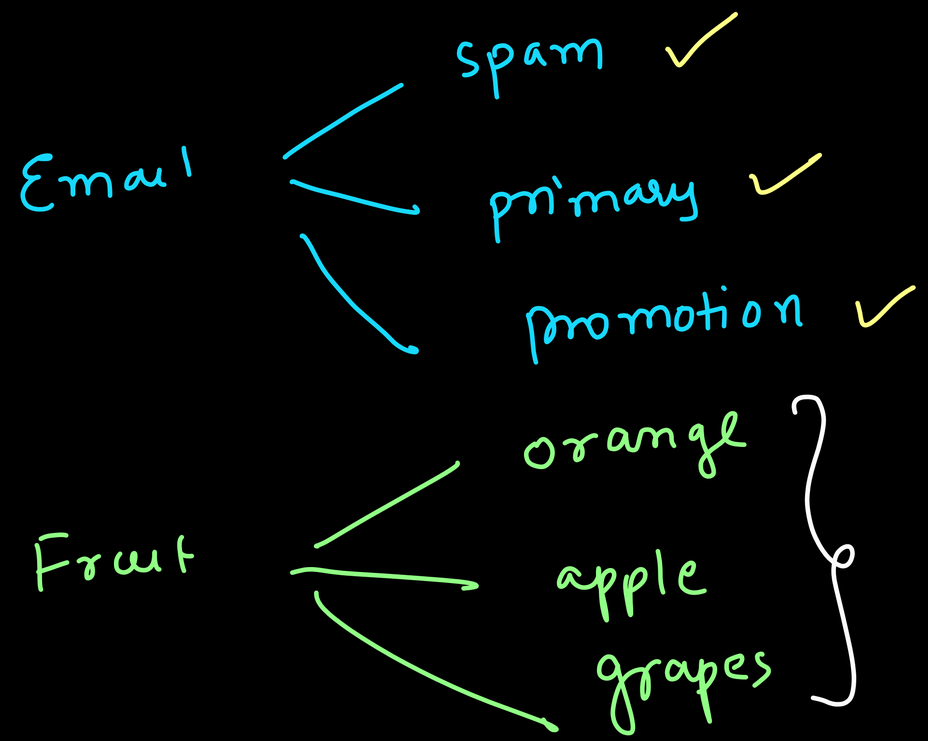
$z = -ve$

$\hat{y} = \sigma(z) \simeq 0.000001$

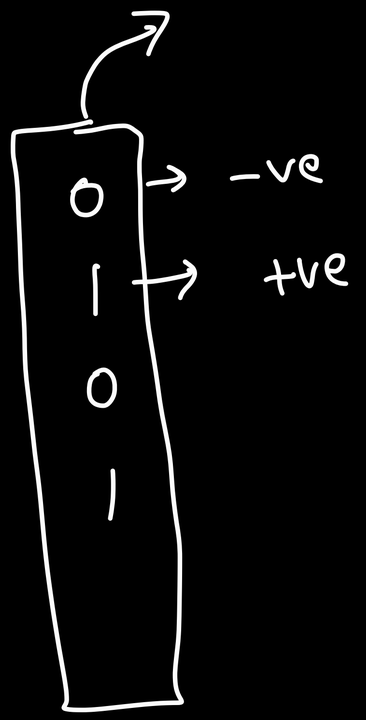
$L = -(-\infty) = +\infty$



Multi class ≠ works for Binary

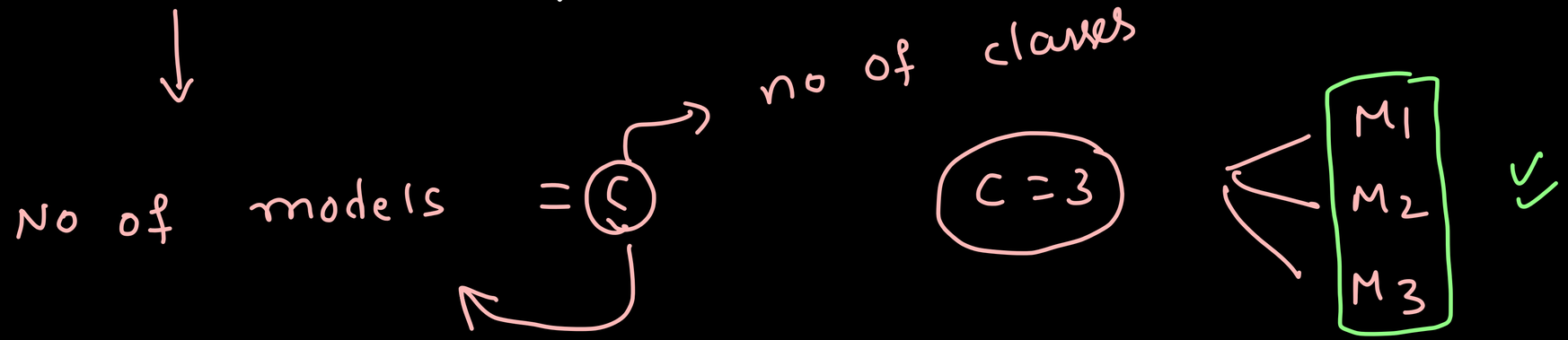


0
1
2

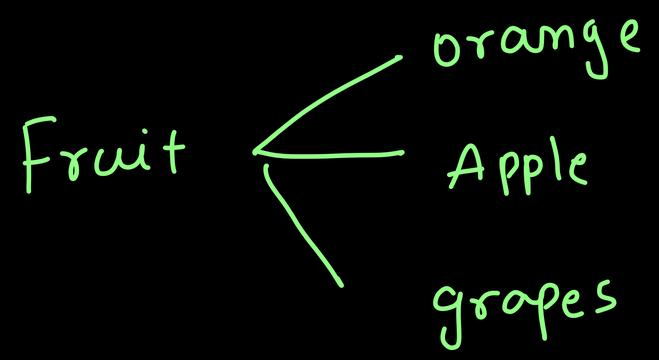


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Technique → one vs All / one vs Rest



-

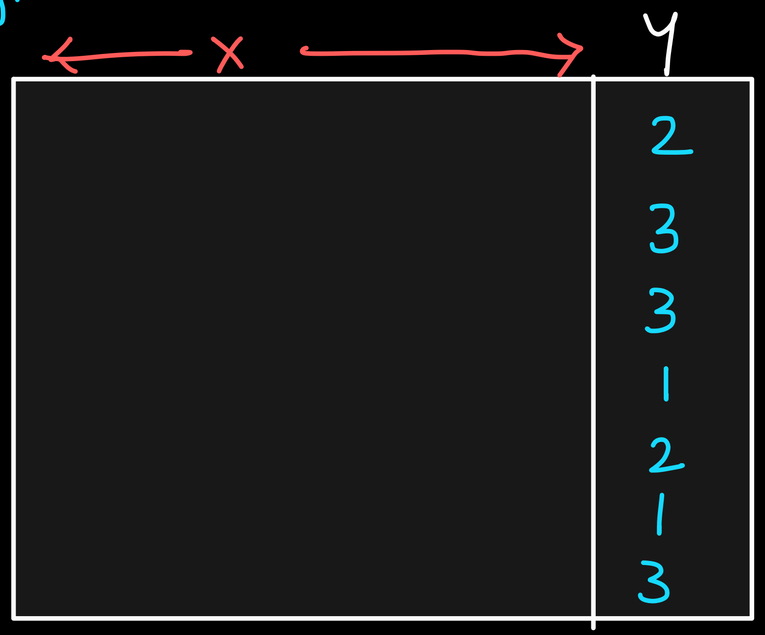


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original data

What is the purpose of M_1 ⟹ orange (or) not

1 0



1 → orange

2 → Apple

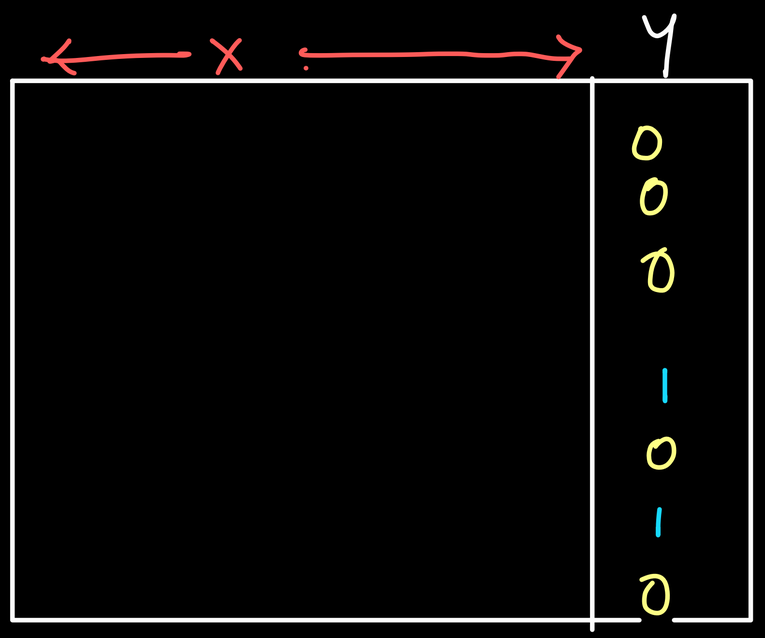
3 → grapes

M2 → Apple (or) not

M3 → grapes (or) not

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For M_1 data



→

orange

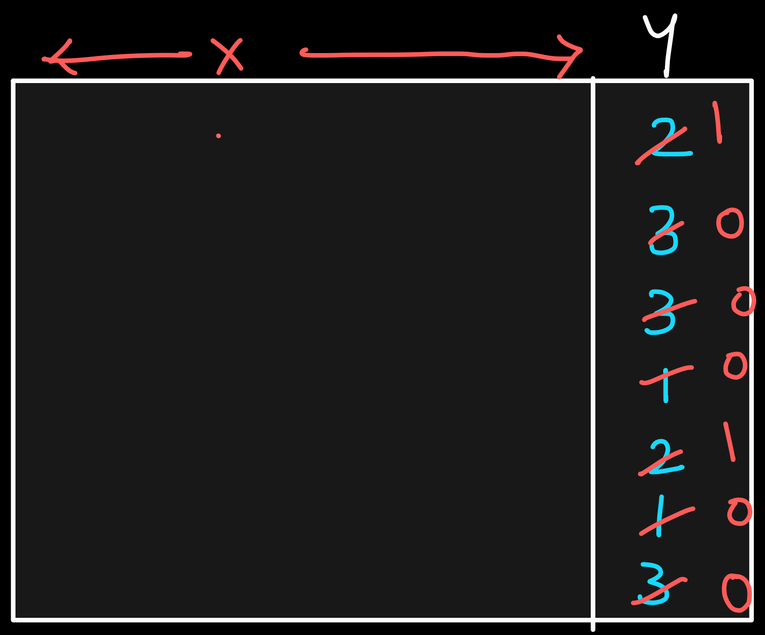
↑

class 1 → 1 ✓

else → 0

M_1 | train log reg-1 ⇒ weather

For M_2 data

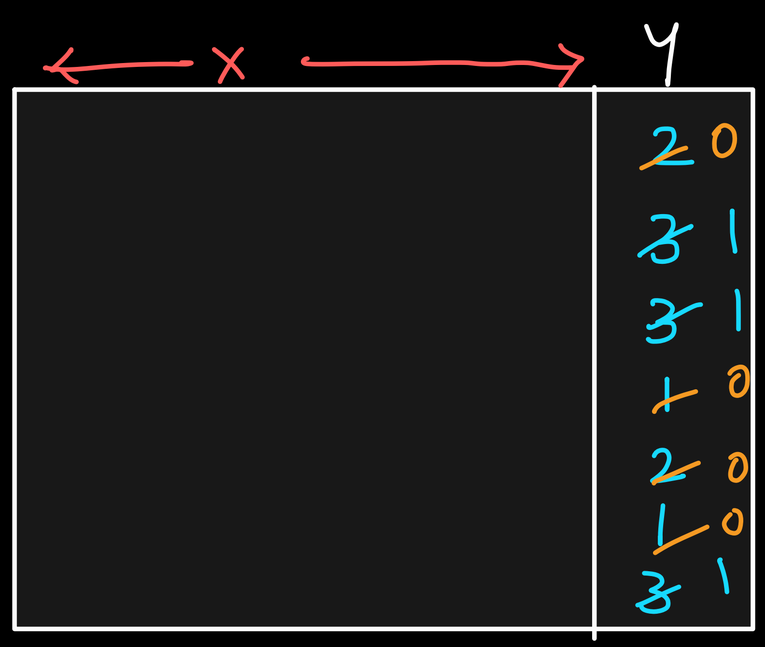


apple ↑

class 2 → 1

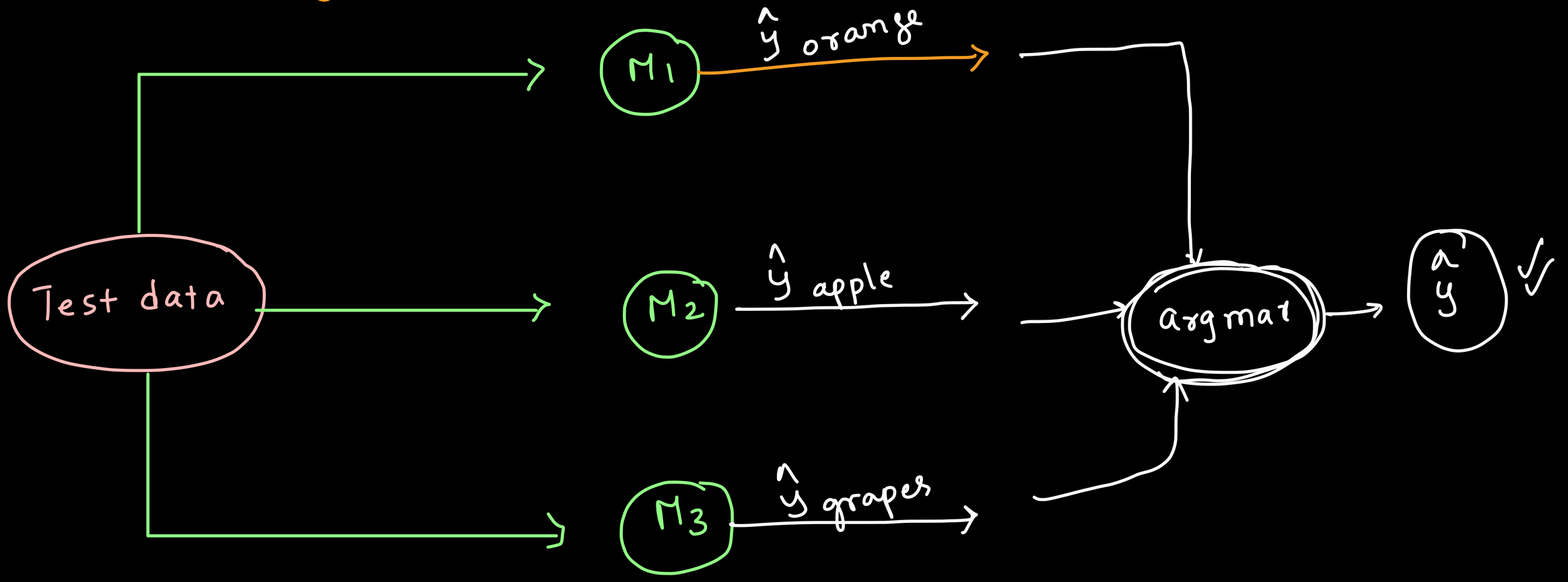
else → 0

M3



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How do you make prediction ?



$M_1 \rightarrow 0.86$

$M_2 \rightarrow 0.05$

$M_3 \rightarrow 0.09$

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Quiz

What is the purpose of the one-vs-rest (OvR) strategy in multi-class logistic regression?

0 users have participated

A	To improve the interpretability of the model coefficients	0%
B	To handle imbalanced datasets in multi-class problems	0%
C	To reduce the complexity of the model	0%
D	To transform a multi-class problem into multiple binary classification problems	0%

End Quiz Now

c = 3

3

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